

Grade 6 Accelerated
Unit 13
Lesson 1 Practice

Name Answer Key
Date _____
Period _____

1. What is the value of the expression $6 \cdot 2 - (x - 4)$ if $x = 1$?

$$\begin{aligned} &6 \cdot 2 - (1 - 4) \\ &12 - 1 + 4 \\ &11 + 4 \\ &15 \end{aligned}$$

2. Find the value of $8 - 2(4 + y) + 2y \cdot 2$ when $y = 3$.

$$\begin{aligned} &8 - 2(4 + 3) + 2(3) \cdot 2 \\ &8 - 8 - 6 + 6 \cdot 2 \\ &\cancel{8} - \cancel{8} - 6 + 12 \\ &6 \end{aligned}$$

3. Evaluate $2p + \frac{16}{j} - 11$ for $p = 5$ and $j = 8$.

$$\begin{aligned} &2(5) + \frac{16}{8} - 11 \\ &10 + 2 - 11 \\ &12 - 11 \\ &1 \end{aligned}$$

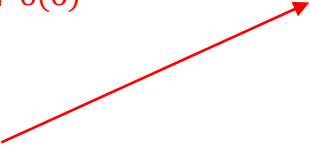
4. Evaluate $10 - 20 \div 4f + 11$ if $f = 2$.

$$\begin{aligned} &10 - 20 \div 4(2) + 11 \\ &10 - 5(2) + 11 \\ &\cancel{10} - \cancel{10} + 11 \\ &11 \end{aligned}$$

5. Find the value of $7 \cdot (2m - 4m + 18)$ when $m = 3$.

$$\begin{aligned} &7(2 \cdot 3 - 4 \cdot 3 + 18) \\ &7(6 - 12 + 18) \\ &7(-6 + 18) \\ &7(12) \\ &84 \end{aligned}$$

6. Find the value of $7 - (y + 12)^2 + 6m$ if $m = 6$ and $y = 2$.

$$\begin{aligned} &= 7 - (2 + 12)^2 + 6(6) &= -189 + 36 \\ &= 7 - (14)^2 + 36 &= -153 \\ &= 7 - 196 + 36 \end{aligned}$$


7. Find the value of $12g - (16 - u - 8) \div 2$ if $g = 3$ and $u = 11$.

$$\begin{aligned} &12(3) - (16 - 11 - 8) \div 2 \\ &12(3) + (-16 + 11 + 8) \div 2 \\ &12(3) + (-5 + 8) \div 2 \\ &12(3) + (3) \div 2 \\ &36 + 3 \div 2 \\ &36 + 1.5 \\ &37.5 \end{aligned}$$

8. Evaluate $8a + 4b(b + 10)$ for $a = 3$ and $b = 2$.

$$\begin{aligned} &8(3) + 4(2)(2 + 10) \\ &8(3) + 4(2)(12) \\ &24 + 8(12) \\ &24 + 96 \\ &120 \end{aligned}$$

9. What is the value of the expression $5g \cdot h^3 + (h - 1)^2$ if $g = 1$ and $h = 2$?

$$\begin{aligned} &= 5(1) \cdot 2^3 + (2 - 1)^2 \\ &= 5 \cdot 8 + (1)^2 \\ &= 40 + 1 \\ &= 41 \end{aligned}$$

10. What is the value of the expression $\left(\frac{b^2-6}{4}\right) \cdot 7 + y^2$ if $b = 2$ and $y = 5$?

$$\begin{aligned} &= \left(\frac{2^2-6}{4}\right) \cdot 7 + 5^2 &= -\frac{1}{2} \cdot \frac{7}{1} + 25 \\ &= \left(\frac{4-6}{4}\right) \cdot 7 + 25 &= -3.5 + 25 \\ &= \left(\frac{-2}{4}\right) \cdot 7 + 25 &= 21.5 \end{aligned}$$
